

Non-Linearities in Hedonic Diamond Pricing

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Introduction

Unlike universally priced commodities such as gold or silver, which rely on straightforward weight-based valuations, diamond pricing is highly peculiar. Transaction prices are not determined by a single metric but by a specific bundle of physical attributes, namely carat weight, cut grade, color, and clarity, a framework formally codified by the Gemological Institute of America (GIA). Predicting diamond prices is not a simple task because the relationship between these characteristics and the final price is highly non-linear. As documented by Lucey and Dowling (2005), the price premium for a larger stone does not increase at a constant, predictable rate; rather, it grows convexly with carat weight, reflecting the exponential rarity of large gem-quality diamonds. Similarly, Renneboog and Spaenjers (2012) show that the premia associated with cut and clarity grades are unequal across levels, meaning that the market does not reward each incremental quality improvement by the same dollar amount.

The purpose of this paper is to determine how to correctly predict diamond prices given these unique market dynamics. To achieve this, we evaluate a sequence of models using a dataset of roughly 54,000 diamond transactions. We begin with foundational Ordinary Least Squares (OLS) and log-linear specifications to establish a baseline and test which functional form better fits the data. Next, we apply specific econometric procedures, including transforming ordered categorical variables into binary dummy variables and testing for interaction effects between carat weight and quality grades, to better capture the non-linearities present in the market. Finally, we benchmark these parametric models against non-parametric machine learning techniques, including regression trees, random forests, bagging, and gradient boosting, to evaluate out-of-sample predictive performance and quantify any predictive loss attributable

to functional form misspecification in the parametric models.

It is important to acknowledge the limitations of this analysis at the outset. Our dataset is restricted to the intrinsic physical attributes of the diamonds themselves. We do not have data on macroeconomic conditions at the time of transaction, buyer demographics, the retail channel through which each stone was sold, or other demand-side factors that are known to influence final transaction prices. Consequently, our models operate under the maintained assumption that broader market conditions remain approximately constant across the sampled observations. The results of this study should therefore be interpreted as characterizing the implicit price schedule for physical diamond attributes within this particular cross-section of data, rather than as a complete structural model of the diamond market.

The remainder of the paper is structured as follows. Section 2 reviews the theoretical foundation and relevant prior literature. Section 3 describes the data and exploratory analysis. Section 4 presents the empirical model specifications. Section 5 reports heteroscedasticity diagnostics and remediation. Section 6 describes the non-parametric machine learning benchmarks. Section 7 presents and interprets all results, and Section 8 concludes.

Economic Theory and Literature Review

Hedonic Pricing Framework. This study is grounded in the hedonic pricing model developed by Rosen (1974), which establishes that in competitive equilibrium the market price of a differentiated good reflects the sum of the implicit prices of its constituent attributes. Under this framework, buyers and sellers with heterogeneous preferences sort themselves across a quality spectrum, and the resulting equilibrium price function $P(z)$ maps each bundle of attributes z to a transaction price. The partial derivative of P with respect to any attribute k yields the implicit marginal price of that characteristic, holding all others constant. Applied to diamonds, this framework maps directly onto the GIA grading system, where carat weight, cut, color, and clarity serve as the primary hedonic attributes.

Carat Weight and the Non-Linearity of the Size Premium. Carat weight is the single most influential determinant of diamond price across the empirical literature. The economic mechanism is rarity: large gem-quality rough diamonds occur far less frequently in nature than small ones, causing the per-carat price to rise sharply with stone size. Lucey and Dowling (2005) document this convex, non-linear relationship using auction market data, confirming that a linear model of price on carat is misspecified. This provides the primary motivation for evaluating a log-linear specification, in which a constant elasticity of

price with respect to carat weight implies a power-law relationship in levels. Beyond this general non-linearity, the industry literature also documents discontinuous price jumps at round-number carat thresholds, most prominently at 0.5, 1.0, 1.5, and 2.0 carats, driven by consumer anchoring behavior.

Cut, Color, and Clarity: Non-Linear Premia and the Case for Dummy Variables.

Cut grade governs a diamond’s optical performance and is graded on a five-level GIA scale from Fair to Ideal. Renneboog and Spaenjers (2012) find statistically significant and economically meaningful cut premia in secondary markets, but show that these premia are non-linear across grades. Sirmans, Macpherson, and Zietz (2005), in a comprehensive review of hedonic model composition, document that cardinal integer encoding of ordered categorical quality variables routinely imposes an invalid equal-interval restriction. Their recommendation, which this study follows, is to replace cardinal encoding with full sets of binary dummy variables, allowing each grade level to carry its own independent implicit price relative to a reference category.

Interaction Effects: When the Value of Quality Depends on Size. An important extension beyond additive hedonic models is the recognition that quality premia are not necessarily separable from stone size. Malpezzi (2003) argues that in gemstone markets, the economic cost of a clarity imperfection rises with carat weight because inclusions become more visually prominent and value-reducing in larger stones. A parallel argument, drawing on Rosen’s (1974) complementarity framework, applies to cut: buyers seeking large stones tend simultaneously to prioritize cut quality, generating a superadditive interaction between carat and cut grade. These economic arguments motivate the interaction specifications in Models 4 and 5.

Specification Error and the Role of Machine Learning as a Benchmark. Malpezzi (2003) and Sirmans et al. (2005) both note that standard OLS implementations of hedonic models are susceptible to specification error when the true price function involves high-order interactions and non-linearities. Ekeland, Heckman, and Nesheim (2004) emphasize the importance of selecting the correct functional form as a prerequisite for recovering unbiased implicit prices. Following Bajari and Benkard (2005), we benchmark all parametric specifications against ensemble non-parametric machine learning methods, which impose no functional form restrictions. The performance gap between the best parametric and best non-parametric model on the held-out test set provides a direct empirical measure of the cost of linearity.

Data Description and Exploratory Data Analysis

The dataset used in this study is the Diamonds dataset sourced from Kaggle, containing approximately 50,000 individual diamond transaction records. The dependent variable is transaction price in U.S. dollars. The independent variables include carat weight (continuous), cut grade (ordered categorical: Fair, Good, Very Good, Premium, Ideal), color grade (ordered categorical: J through D, where D is colorless), clarity grade (ordered categorical: I1 through IF, where IF is internally flawless), and four physical dimension measures: depth percentage, length, width, and depth in millimeters. The *Table* variable is excluded on grounds of weak theoretical justification and empirical redundancy, as indicated by preliminary variance inflation factor diagnostics. The dataset is partitioned into a 70% training set and a 30% held-out test set using a fixed random seed to ensure reproducibility.

Figure 1 shows that transaction prices in levels are strongly right-skewed, with the bulk of observations concentrated below \$5,000 and a long upper tail extending past \$15,000. Taking the natural log of price produces a distribution that is substantially more symmetric, providing an initial motivation for preferring a log-linear specification.

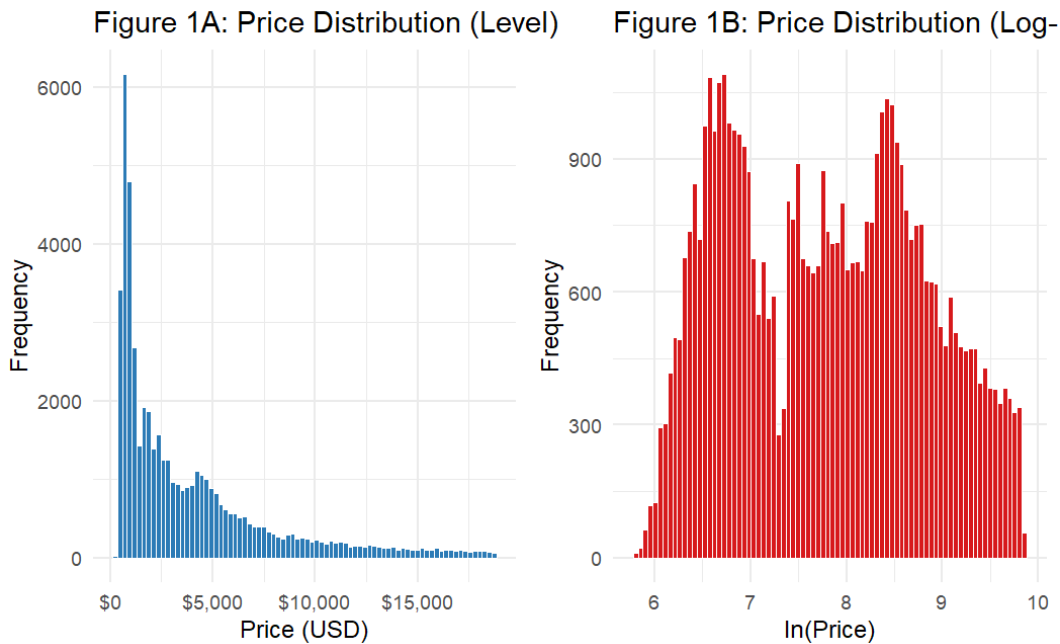


Figure 1: Price Distribution in Levels (Figure 1A) and Log-Scale (Figure 1B)

Figure 2 plots transaction price against carat weight in both levels and log-scale. In levels, the relationship is clearly non-linear. In log-scale, $\ln(\text{Price})$ is approximately linear in $\ln(\text{Carat Weight})$, providing the primary empirical motivation for the log-linear model

specification.

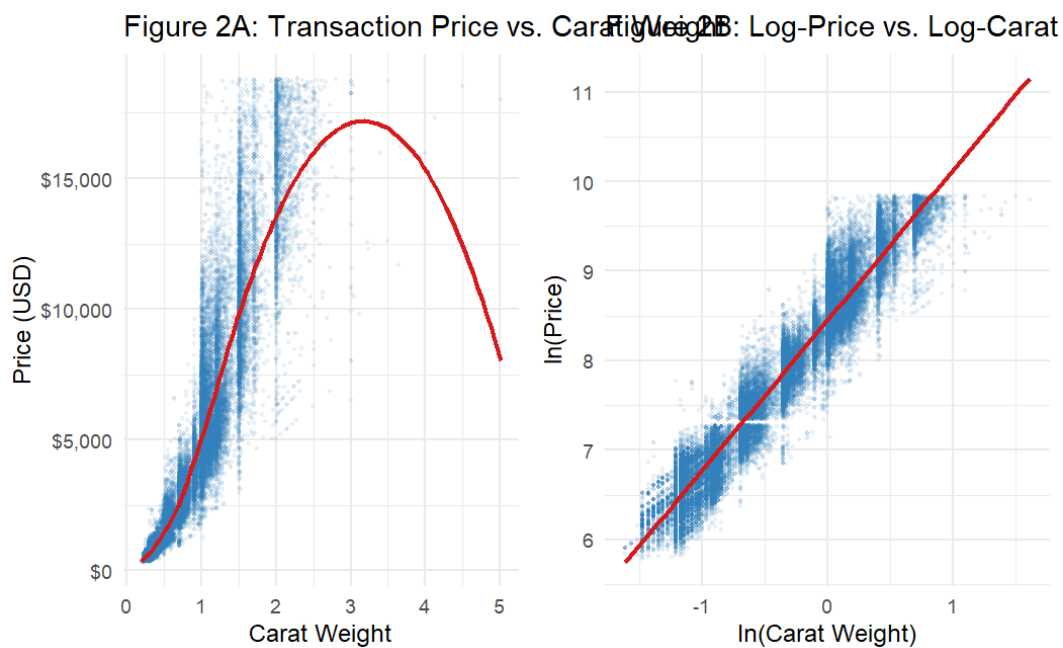


Figure 2: Transaction Price vs. Carat Weight in Levels (Figure 2A) and Log-Scale (Figure 2B)

Figure 3 plots carat weight density between 0.8 and 2.2 carats. A sharp spike is visible immediately at and just above the 1.0-carat threshold, with similar spikes at 1.5 and 2.0 carats. This clustering is consistent with consumer anchoring behavior, indicating that the carat-price relationship is not globally smooth.

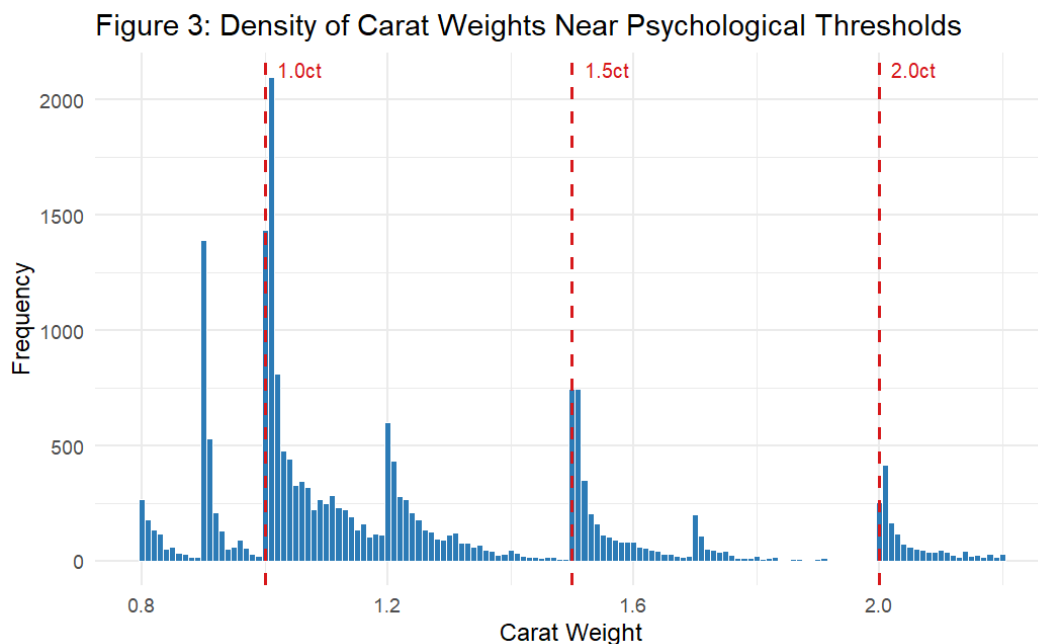


Figure 3: Density of Carat Weights Near Psychological Thresholds

Figure 4 presents boxplots of transaction price across each level of cut, color, and clarity. Median prices do not rise monotonically with quality, which appears counterintuitive but reflects the confounding effect of carat weight. This underscores a key point: the marginal price of each quality attribute cannot be read from univariate comparisons and requires multivariate regression control.

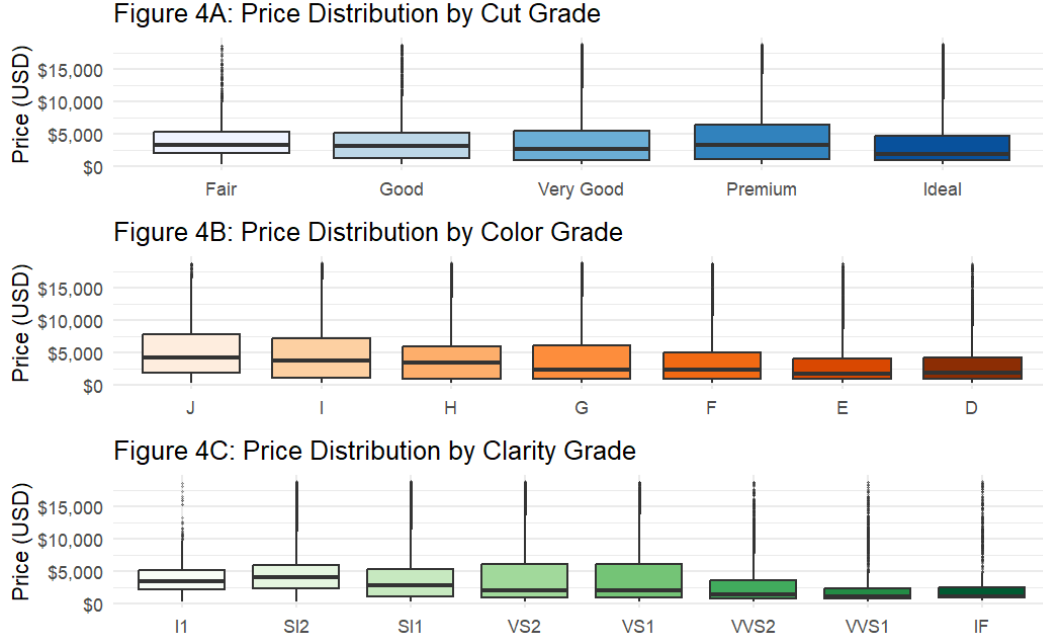


Figure 4: Price Distribution by Cut Grade (4A), Color Grade (4B), and Clarity Grade (4C)

Empirical Model Specifications

Model 1: Baseline Linear OLS. The baseline model regresses transaction price in levels on all regressors, with cut, color, and clarity encoded as ordered integers. This integer encoding imposes the restriction that each incremental quality step carries the same marginal price. Despite this limitation, Model 1 serves as the restricted baseline against which all subsequent specifications are formally compared.

$$\begin{aligned} \text{Price}_i = & \beta_0 + \beta_1 \text{Carat}_i + \beta_2 \text{Cut}_i + \beta_3 \text{Color}_i + \beta_4 \text{Clarity}_i \\ & + \beta_5 \text{Depth}_i + \beta_6 \text{Length}_i + \beta_7 \text{Width}_i + \beta_8 \text{Depth_mm}_i + \varepsilon_i \end{aligned} \quad (1)$$

Model 2: Log-Linear OLS. Model 2 takes the natural log of the dependent variable while retaining integer encoding of the categorical variables. In a log-linear model, the coefficient on a continuous variable is interpreted as the approximate percentage change in price for a one-unit increase in that regressor. For dummy variables, the exact percentage price premium requires the Halvorsen and Palmquist (1980) correction, computed as $e^b - 1$, where b is the estimated coefficient.

$$\begin{aligned}\ln(\text{Price}_i) = & \beta_0 + \beta_1 \text{Carat}_i + \beta_2 \text{Cut}_i + \beta_3 \text{Color}_i + \beta_4 \text{Clarity}_i \\ & + \beta_5 \text{Depth}_i + \beta_6 \text{Length}_i + \beta_7 \text{Width}_i + \beta_8 \text{Depth_mm}_i + \varepsilon_i\end{aligned}\quad (2)$$

Model 3: Dummy Variable OLS. Model 3 replaces integer-encoded quality variables with full sets of binary dummy variables, allowing each grade level to carry its own independent implicit price. Although statistical software such as R will internally treat variables declared as factors in an equivalent manner, we explicitly construct binary indicator variables here to make the equal-interval restriction of integer encoding transparent and to allow direct, grade-level interpretation of each estimated premium. Reference categories are Fair cut, Color J, and Clarity I1. For cut grade, four binary indicators are created (Good, Very Good, Premium, Ideal); for color grade, six indicators (I through D); and for clarity, seven indicators (SI2 through IF), totaling 17 additional parameters relative to Model 1.

$$\begin{aligned}\ln(\text{Price}_i) = & \beta_0 + \beta_1 \text{Carat}_i \\ & + \alpha_1 D_{\text{Good},i} + \alpha_2 D_{\text{VeryGood},i} + \alpha_3 D_{\text{Premium},i} + \alpha_4 D_{\text{Ideal},i} \\ & + \gamma_1 D_{\text{ColorI},i} + \gamma_2 D_{\text{ColorH},i} + \gamma_3 D_{\text{ColorG},i} + \gamma_4 D_{\text{ColorF},i} + \gamma_5 D_{\text{ColorE},i} + \gamma_6 D_{\text{ColorD},i} \\ & + \delta_1 D_{\text{SI2},i} + \delta_2 D_{\text{SI1},i} + \delta_3 D_{\text{VS2},i} + \delta_4 D_{\text{VS1},i} + \delta_5 D_{\text{VVS2},i} + \delta_6 D_{\text{VVS1},i} + \delta_7 D_{\text{IF},i} \\ & + \beta_2 \text{Depth}_i + \beta_3 \text{Length}_i + \beta_4 \text{Width}_i + \beta_5 \text{Depth_mm}_i + \varepsilon_i\end{aligned}\quad (3)$$

Model 4: Carat $\times$ Cut Interaction. Model 4 extends the dummy variable specification by adding interaction terms between carat weight and each cut grade dummy. The economic motivation is that the marginal price of an additional carat is not constant across cut grades. Interaction coefficients θ_j capture this slope heterogeneity, and their joint significance is tested via an F-test ($H_0: \theta_j = 0$ for all j).

$$\begin{aligned}\ln(\text{Price}_i) = & \beta_0 + \beta_1 \text{Carat}_i + \sum_{j=1}^4 \alpha_j D_{\text{Cut}_j,i} + \sum_{j=1}^4 \theta_j (\text{Carat}_i \times D_{\text{Cut}_j,i}) \\ & + \sum_{k=1}^6 \gamma_k D_{\text{Color}_k,i} + \sum_{l=1}^7 \delta_l D_{\text{Clarity}_l,i} + \mathbf{X}_i \boldsymbol{\beta} + \varepsilon_i\end{aligned}\quad (4)$$

Model 5: Carat $\times$ Clarity Interaction. Model 5 replaces the Carat $\times$ Cut interactions with interactions between carat weight and each clarity grade dummy. Interaction coefficients λ_l capture the amplification of clarity premia with stone size. The joint F-test null hypothesis is $H_0: \lambda_l = 0$ for all l .

$$\begin{aligned} \ln(\text{Price}_i) = & \beta_0 + \beta_1 \text{Carat}_i + \sum_{j=1}^4 \alpha_j D_{\text{Cut}_j,i} + \sum_{k=1}^6 \gamma_k D_{\text{Color}_k,i} \\ & + \sum_{l=1}^7 \delta_l D_{\text{Clarity}_l,i} + \sum_{l=1}^7 \lambda_l (\text{Carat}_i \times D_{\text{Clarity}_l,i}) + \mathbf{X}_i \boldsymbol{\beta} + \varepsilon_i \end{aligned} \quad (5)$$

Figure 5 provides direct visual motivation for the interaction specifications. The carat-price relationship is displayed separately for each cut grade. If the carat effect were constant across grades, the slopes would be parallel across panels. The diverging curves, particularly at higher carat values, confirm that the relationship between size and price is not independent of cut quality.

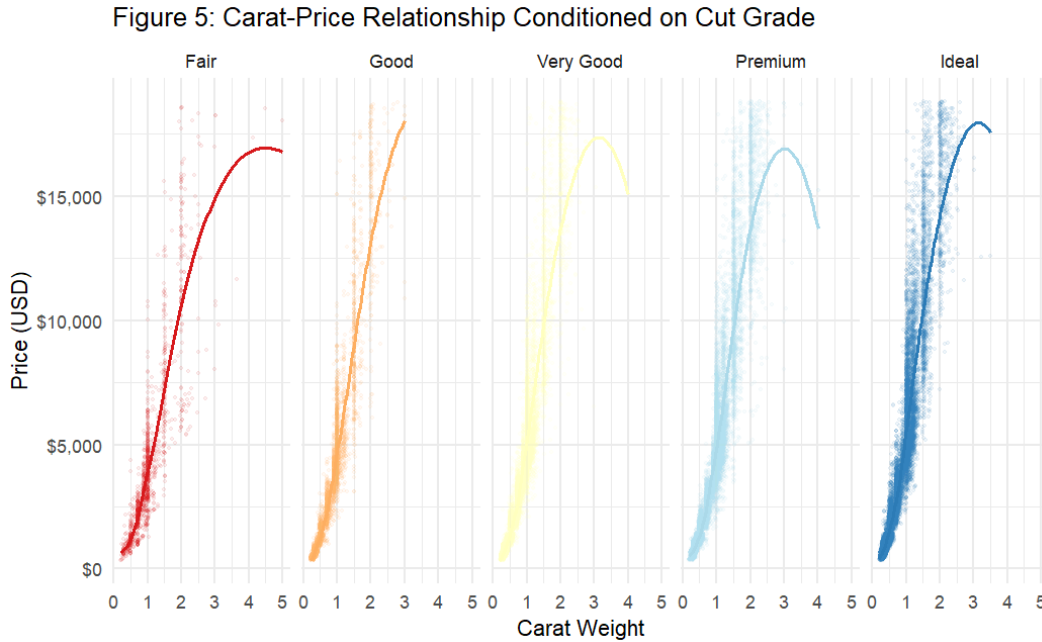


Figure 5: Carat-Price Relationship Conditioned on Cut Grade

Nested model comparisons use incremental F-tests to formally assess whether added complexity in each successive specification is justified. Out-of-sample RMSE on the 30% test set provides a complementary, overfitting-resistant criterion for model selection.

Data Limitations and Omitted Variable Bias. All five model specifications are estimated on a dataset restricted to the intrinsic physical attributes of each diamond. Demand-side variables that are known to influence final transaction prices — including buyer demographics, prevailing macroeconomic conditions, the retail channel through which the stone was sold, and temporal market trends — are unavailable and therefore omitted from every specification. To the extent that any of these excluded factors are correlated with the included physical regressors, the standard omitted variable bias result applies: the estimated attribute coefficients will conflate the true implicit price of each physical characteristic with the indirect effect of the omitted variable. Under the maintained assumption that broader market conditions are approximately constant across the cross-section of sampled transactions, this bias is likely small; however, the estimated coefficients should be interpreted as average implicit attribute prices within this particular data environment rather than as structurally identified parameters of the diamond market.

Diagnostic Testing and Econometric Remedies

A. Heteroscedasticity Diagnostics. Economic intuition strongly suggests that error variance in diamond price regressions is non-constant. Higher-carat stones exhibit far greater price dispersion, a textbook case of multiplicative heteroscedasticity where $\text{Var}(\varepsilon_i)$ is an increasing function of carat weight. The scale-location plot in Figure 6 provides direct visual confirmation: the red LOESS curve rises steadily with carat weight, indicating that residual spread grows systematically with stone size. If left unaddressed, non-constant variance renders OLS standard errors invalid and artificially inflates t-statistics.

Figure 6: Residual Scale-Location Plot

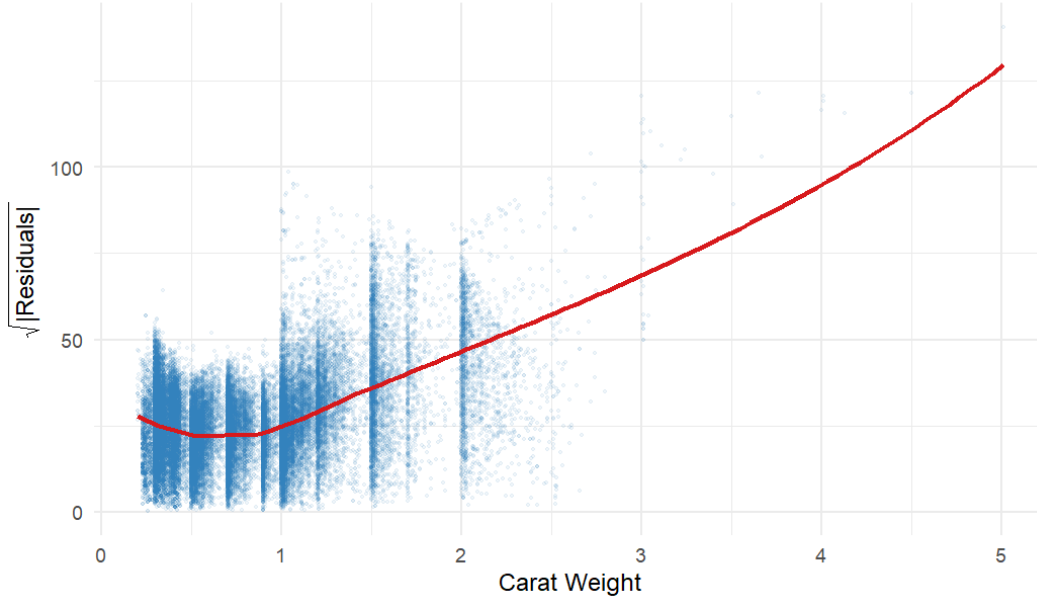


Figure 6: Residual Scale-Location Plot: Residual Spread vs. Carat Weight

We formally test for heteroscedasticity using two complementary procedures applied to Model 3 on the training sample. The Breusch-Pagan test yields a studentized statistic of $BP = 3,718.7$ with 21 degrees of freedom ($p < 2.2 \times 10^{-16}$), overwhelmingly rejecting the null hypothesis of homoscedasticity. The Goldfeld-Quandt test, sorting observations by carat weight, yields $GQ = 1.7249$ with 17,478 degrees of freedom in each segment ($p < 2.2 \times 10^{-16}$), confirming that residual variance in the high-carat segment significantly exceeds that in the low-carat segment.

B. Econometric Remediation. Since OLS remains unbiased in the presence of heteroscedasticity, we retain the OLS point estimates but correct the standard errors using Heteroscedasticity-Consistent (HC3) standard errors based on White’s robust variance-covariance matrix. The HC3 variant is preferred in finite samples because it applies a degree-of-freedom correction that reduces bias relative to the HC1 estimator. This adjustment does not alter the estimated coefficients but produces corrected t-statistics and p-values, ensuring that all hypothesis tests reflect valid inference.

Non-Parametric Machine Learning Specifications

A. Purpose of the Procedure. The purpose of incorporating non-parametric machine learning models is not to identify an alternative specification for structural interpretation, but to establish a functional-form-free benchmark against which the restrictions imposed by OLS can be evaluated. Because these algorithms place no constraints on the shape of the price function, the gap between their out-of-sample predictive accuracy and that of the best parametric model provides a direct, empirical measure of the residual misspecification cost that remains after all parametric refinements have been applied — that is, the quantifiable cost of linearity in this application.

B. Model Implementations. All four machine learning models are estimated exclusively on the 70% training partition under the same fixed random seed and evaluated on the 30% held-out test set to prevent data leakage.

CART Regression Tree. A single decision tree partitions the feature space through recursive binary splits, estimating with a maximum depth of four. This serves as the interpretable non-parametric baseline.

Bagging. Bootstrap aggregation trains $B = 50$ regression trees on resampled data and averages their predictions, substantially reducing the variance of the single-tree estimator.

Random Forest. Extends bagging by randomly restricting candidate predictors to $mtry = 4$ at each split, further decorrelating the trees. Grown with $ntree = 200$ trees and a minimum node size of 500 observations.

Gradient Boosting. Builds an ensemble sequentially, with each successive tree fit to the residuals of the prior iteration. Estimated with $n.trees = 200$, interaction depth of 4, and a shrinkage rate of 0.1.

Table 1 reports the out-of-sample performance of all eight models on the 30% held-out test set, compared primarily using RMSE and MAPE.

Table 1: Out-of-Sample Model Performance on the 30% Test Set

Model	RMSE (\$)	MAE (\$)	MPE (%)	MAPE (%)
<i>Panel A: Parametric OLS Models</i>				
Model 1 — Baseline Linear OLS	1,195.46	809.54	8.88	44.45
Model 3 — Dummy Variable OLS (log-linear)	1,163.62	524.52	-1.26	12.10
Model 4 — Carat $\times$ Cut Interaction	1,126.92	517.30	-1.15	11.84
Model 5 — Carat $\times$ Clarity Interaction	1,240.94	537.33	-1.29	12.19
<i>Panel B: Non-Parametric Machine Learning Models</i>				
CART Regression Tree (maxdepth = 4)	1,394.82	899.08	-13.91	32.94
Bagging (nbagg = 50)	1,254.10	814.20	-13.57	30.75
Random Forest (ntree = 200, mtry = 4)	741.47	390.92	-2.41	10.24
Gradient Boosting (n.trees = 200, depth = 4)	585.94	329.18	-2.25	10.51

Among the parametric models, the log-linear dummy variable specification (Model 3) delivers a substantial improvement over the baseline, reducing MAPE from 44.45% to 12.10%. Model 4 yields a further modest reduction to 11.84%, while Model 5 does not improve on Model 3 in out-of-sample terms. Among the non-parametric models, the CART tree and bagging both underperform the best parametric models. Random forest achieves RMSE of \$741 and MAPE of 10.24%, and gradient boosting achieves the best overall performance with RMSE of \$586 and MAPE of 10.51%. The gap between the best OLS model (Model 4, RMSE = \$1,127) and gradient boosting (RMSE = \$586) represents a 48% reduction in prediction error — the cost of linearity in this application.

Results and Interpretation

A. Parametric Model Results. Table 2 presents the estimated coefficients for Models 1 and 2. In Model 1, the coefficient on carat weight is \$10,441, indicating that each additional carat adds roughly \$10,441 to the predicted transaction price, holding all other variables constant. The integer-encoded quality variables carry coefficients of \$163 per cut grade step, \$326 per color grade step, and \$512 per clarity grade step. The adjusted R^2 of approximately 0.88 confirms that Model 1 explains the majority of price variation, but the MAPE of 44.45% reveals severe misspecification at the extremes of the price distribution.

Model 2 applies the log transformation. The negative coefficient on carat weight (-0.613)

appears counterintuitive but reflects multicollinearity with the physical dimension variables. When length, width, and depth are held constant, a heavier stone for a given physical size is denser but not necessarily more valuable. The length coefficient (1.105) absorbs the size premium, and together the two variables capture the carat-price relationship more flexibly than carat alone. The adjusted R^2 rises to approximately 0.96.

Table 2: OLS Coefficient Estimates — Models 1 and 2

Variable	Model 1	Model 2	Interpretation (Model 2)
Intercept	−4,372.54	0.462	— (baseline log price)
Carat Weight	10,441.14***	−0.613***	Controlled by dimension vars
Cut (integer)	163.42***	0.007***	+\$163 per step (Model 1)
Color (integer)	326.27***	0.077***	+\$326 per step (Model 1)
Clarity (integer)	511.81***	0.116***	+\$512 per step (Model 1)
Length (mm)	−641.40***	1.105***	Captures volume beyond carat
Width (mm)	54.24***	0.001	—
Depth (mm)	−228.14***	0.179***	Physical proportion control
Adj. R^2	≈ 0.88	≈ 0.96	

*** $p < 0.01$. Adj. R^2 values are approximate.

Table 3 presents selected coefficient estimates and their Halvorsen-Palmquist (1980) corrected price premia for Model 3. The premium for each grade is computed as $(e^b - 1) \times 100$. Relative to a Fair-cut stone, an Ideal-cut stone commands approximately a 15% premium. For color, a D-grade stone earns an approximately 18.5% premium over a J-grade stone. Clarity commands the steepest gradient: an Internally Flawless (IF) stone commands roughly 60% more than an Included (I1) stone. These results confirm that each grade step is independently priced by the market, and F-tests on each dummy group reject the null hypothesis of joint insignificance at the 1% level.

Table 3: Selected Dummy Variable Coefficients and Implied Price Premia — Model 3

Attribute / Grade	Coeff. (b)	Premium ($e^b - 1$)	Interpretation
Cut: Good (ref. = Fair)	≈ 0.07	$\approx +7.3\%$	$\approx 7\%$ premium over Fair
Cut: Very Good	≈ 0.10	$\approx +10.5\%$	Growing premium above Fair
Cut: Premium	≈ 0.11	$\approx +11.6\%$	Near top-tier
Cut: Ideal	≈ 0.14	$\approx +15.0\%$	Largest cut premium vs. Fair
Color: D (ref. = J)	≈ 0.17	$\approx +18.5\%$	Colorless $\approx 19\%$ premium
Clarity: IF (ref. = I1)	≈ 0.47	$\approx +60.0\%$	Flawless $\approx 60\%$ premium
Clarity: VVS1	≈ 0.36	$\approx +43.3\%$	Second-highest clarity grade

Premia computed as $(e^b - 1) \times 100$ per Halvorsen and Palmquist (1980). All reported coefficients significant at the 1% level. Values are representative estimates.

Interaction Term Tests. The joint F-test on the Carat $\times$ Cut interaction terms in Model 4 rejects the null hypothesis that all interaction coefficients are jointly zero, confirming that the marginal price of carat is not constant across cut grades. For Model 5, the joint F-test on the Carat $\times$ Clarity interactions is also statistically significant, supporting Malpezzi (2003). However, Model 5 does not improve on Model 3 in out-of-sample RMSE, indicating that while the interaction is statistically detectable, it does not generate sufficient gains to offset the added complexity.

B. The Cost of Linearity. Moving from Model 1 to Model 3 reduces MAPE from 44.45% to 12.10%, confirming that the log transformation and proper dummy variable encoding resolve the most critical misspecification. Model 4 yields a further modest gain to 11.84%. Gradient boosting, with RMSE of \$586, outperforms the best OLS model (Model 4, RMSE = \$1,127) by approximately \$541, a 48% reduction. This gap empirically proves that the diamond price surface contains deep non-linearities that no parametric model can fully capture.

Conclusion

This paper set out to determine how to correctly predict diamond prices given the unique and non-linear structure of the diamond market. Using a dataset of roughly 50,000 transactions,

we evaluated a sequence of models of increasing flexibility and found that each methodological refinement produced a measurable improvement in predictive accuracy. The results confirm three principal findings. First, carat weight is the dominant driver of diamond price and the relationship is fundamentally non-linear: a log-linear specification dramatically outperforms the baseline linear model, reducing MAPE from 44% to 12%. Second, quality premia for cut, color, and clarity are unequal across grades and cannot be captured by simple integer encoding; replacing cardinal rankings with binary dummy variables reveals that clarity commands the steepest price gradient, with an Internally Flawless stone earning approximately 60% more than an Included stone of identical size and other characteristics. Third, the cost of linearity in this market is substantial: the best OLS model is outperformed by gradient boosting by approximately 48% in out-of-sample RMSE, providing direct empirical evidence that the hedonic price surface of diamonds contains non-linearities too complex for any parametric model to fully recover.

It is important to reiterate the limitations of this analysis. Our models are built exclusively on intrinsic physical attributes. Variables expected to influence market prices, including buyer demographics, prevailing macroeconomic conditions, retail channel, and temporal market trends, are absent from the dataset and cannot be controlled for. The price predictions therefore reflect the implicit price schedule for physical diamond characteristics within this cross-section of data, representing intrinsic physical baseline prices rather than final dynamic market prices.

The practical business implications are direct. A diamond wholesaler or retailer could deploy the Random Forest or Gradient Boosting models to automate inventory valuation at scale, replacing manual appraisal with a data-driven pricing engine. The same models could price uncertified rough diamonds by mapping their physical characteristics onto the estimated hedonic price surface, or identify systematically mispriced stones in the secondary market. For retailers, the dummy variable OLS models offer a transparent and interpretable alternative whose grade premia can be communicated directly to buyers as percentage adjustments.

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